

Advanced Mathematical Techniques

Problem Set 4

1. This problem concerns the evaluation of integrals of the form $\int_0^\infty e^{-ax} \ln^n x dx$ for natural numbers n and positive real numbers a .

- (a) Use the integral representation of the Gamma function to write the value of the integral $\int_0^\infty x^\varepsilon e^{-x} dx$.

$$\int_0^\infty x^\varepsilon e^{-x} dx = \boxed{\Gamma(1 + \varepsilon)}$$

- (b) Write the Maclaurin expansion for x^ε , thought of as a function of ε rather than x . Your expansion should contain powers of ε .

$$x^\varepsilon = e^{\varepsilon \ln x} = \boxed{\sum_{k=0}^{\infty} \frac{(\varepsilon \ln x)^k}{k!}}$$

- (c) Substitute your result from part (b) into the integral given in part (a) and exchange the order of summation and integration to obtain a series of the form $\sum_{k=0}^{\infty} c_k \varepsilon^k$, where the coefficients c_k contain integrals involving x . It is not necessary to justify the exchange of summation and integration for this problem.

$$\int_0^\infty x^\varepsilon e^{-x} dx = \int_0^\infty \sum_{k=0}^{\infty} \frac{(\varepsilon \ln x)^k}{k!} e^{-x} dx = \sum_{k=0}^{\infty} \frac{\varepsilon^k}{k!} \int_0^\infty \ln^k(x) e^{-x} dx$$

$$c_k = \boxed{\frac{1}{k!} \int_0^\infty \ln^k(x) e^{-x} dx}$$

- (d) Use the Maclaurin expansion for $\ln \Gamma(1+x)$ derived in class to determine the first four nonzero terms in the Maclaurin expansion for $\Gamma(1+\varepsilon)$, thought of as a function of ε . Your coefficients may be left in terms of the Zeta function at integer arguments and/or the Euler-Mascheroni constant γ .

$$\begin{aligned} \ln \Gamma(x) &= -\gamma\varepsilon + \frac{\zeta(2)}{2}\varepsilon^2 - \frac{\zeta(3)}{3}\varepsilon^3 + \frac{\zeta(4)}{4}\varepsilon^4 - \dots \\ \Gamma(x) &= e^{-\gamma\varepsilon + \frac{\zeta(2)}{2}\varepsilon^2 - \frac{\zeta(3)}{3}\varepsilon^3 + \frac{\zeta(4)}{4}\varepsilon^4 - \dots} \\ &= 1 + (-\gamma\varepsilon + \frac{\zeta(2)}{2}\varepsilon^2 - \frac{\zeta(3)}{3}\varepsilon^3 + \frac{\zeta(4)}{4}\varepsilon^4 - \dots) + \frac{1}{2!}(-\gamma\varepsilon + \frac{\zeta(2)}{2}\varepsilon^2 - \frac{\zeta(3)}{3}\varepsilon^3 + \frac{\zeta(4)}{4}\varepsilon^4 - \dots)^2 + \frac{1}{3!} \dots \end{aligned}$$

Of this, the first 4 terms are:

$$\boxed{1 - \gamma\varepsilon + \varepsilon^2 \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) - \varepsilon^3 \left(\frac{\zeta(3)}{3} + \frac{\pi^2\gamma}{12} + \frac{\gamma^3}{6} \right)}$$

- (e) Combine your results from parts (a), (c), and (d) to determine the values of the following integrals:

By using the coefficients found above,

- i. $\int_0^\infty e^{-x} \ln x dx = \boxed{-\gamma}$
- ii. $\int_0^\infty e^{-x} \ln^2 x dx = \boxed{\zeta(2) + \gamma^2}$
- iii. $\int_0^\infty e^{-x} \ln^3 x dx = \boxed{-2\zeta(3) - 3\gamma\zeta(2) - \gamma^3}$

- (f) Redo part (a) for the integral $\int_0^\infty x^\varepsilon e^{-ax} dx$, assuming a is a positive real number. Then, use your result from part (d) to find the first four nonzero terms in a Maclaurin series for this integral in powers of ε . Hint: This can be done very easily with a few observations. Note that $7^\varepsilon = e^{\varepsilon \ln 7}$, so $7^\varepsilon e^{4\varepsilon} = e^{(4+\ln 7)\varepsilon}$.

$$u = ax, x = \frac{u}{a}, dx = \frac{du}{a}, \int_0^\infty \left(\frac{u}{a}\right)^\varepsilon e^{-u} \frac{du}{a} = a^{-1-\varepsilon} \Gamma(1+\varepsilon) = \frac{1}{a} \exp\left(-(\gamma + \ln a)\varepsilon + \frac{\zeta(2)}{2}\varepsilon^2 - \frac{\zeta(3)}{3}\varepsilon^3 + \dots\right)$$

$$= \frac{1}{a} \left[1 - (\gamma + \ln a)\varepsilon + \left(\frac{\pi^2}{12} + \frac{(\gamma + \ln a)^2}{2}\right)\varepsilon^2 - \left(\frac{\pi^2(\gamma + \ln a)}{12} \frac{2\zeta(3) + (\gamma + \ln a)^3}{6}\varepsilon^3 + \dots\right) \right]$$

- (g) Argue that there is no substantive difference between performing the manipulation in part (c) for $\int_0^\infty x^\varepsilon e^{-x}$ and performing that manipulation for $\int_0^\infty x^\varepsilon e^{-ax}$. Then, use your result from part (f) in conjunction with this fact to evaluate $\int_0^\infty e^{-5x} \ln^2 x dx$.

There is no substantive difference because we see that regardless of the expansion of the e^{-x} term remains independent of the sum.

$$a = 5, \varepsilon = 2, c_2 = \frac{1}{5} \left(\frac{\zeta(2) + (\gamma + \ln 5)^2}{2} \right)$$

$$\int_0^\infty e^{-5x} \ln^2 x dx = 2! \cdot c_2 = \frac{1}{5} \left(\frac{\pi^2}{6} + (\gamma + \ln 5)^2 \right)$$

2. This problem concerns the evaluation of the integral $\int_0^\infty x e^{-3x} \ln^2 x dx$.

- (a) Evaluate the integral $\int_0^\infty x^{1+\varepsilon} e^{-3x} dx$ in terms of the Gamma function.

$$u = 3x, x = \frac{u}{3}, dx = \frac{du}{3}, \int_0^\infty x^{1+\varepsilon} e^{-3x} dx = \boxed{3^{-2-\varepsilon} \Gamma(2+\varepsilon)}$$

- (b) Use the recurrence relation for the Gamma function to rewrite your result from part (a) in a form that allows you to make use of the expansion found in part (d) of problem 1.

$$3^{-2-\varepsilon} \Gamma(2+\varepsilon) = 3^{-2-\varepsilon} (1+\varepsilon) \Gamma(1+\varepsilon) = \boxed{\frac{(1+\varepsilon)}{9} e^{-\varepsilon \ln 3} \Gamma(1+\varepsilon)}$$

- (c) Write out the first 3 nonzero terms of the Maclaurin expansion for the integral you found in part (a).

$$\frac{1}{9} (1 - \varepsilon \ln 3 + \frac{\varepsilon^2 \ln^2 3}{2!} + \dots) \left(1 - \gamma \varepsilon + \frac{\gamma^2}{2} \varepsilon^2 + \frac{\pi^2}{12} \varepsilon^2 - \dots \right)$$

$$= \frac{1}{9} + \frac{\varepsilon(1 - \ln 3 - \gamma)}{9} + \frac{\varepsilon^2 (\frac{\ln^2 3}{2} + \frac{\gamma^2}{2} + \frac{\pi^2}{12} + \gamma \ln 3 - \gamma - \ln 3)}{9}$$

$$= \boxed{\frac{1}{9} \left[1 + (1 - \ln 3 - \gamma)\varepsilon + \left(\frac{\pi^2}{12} + \frac{(\gamma + \ln 3)^2}{2} - \gamma - \ln 3 \right) \varepsilon^2 \right]}$$

- (d) Use the manipulations you performed in parts (b) and (c) of problem 1 to write a Maclaurin expansion of the form $\sum_0^\infty c_k \varepsilon^k$ for the integral in part (a). Your coefficients will contain integrals involving x . Note that you do *not* want powers of $(1+\varepsilon)$. You want powers of ε .

$$\int_0^\infty x^{1+\varepsilon} e^{-3x} dx = \int_0^\infty x \left(\sum_{k=0}^\infty \frac{\ln^k x}{k!} \varepsilon^k \right) e^{-3x} dx = \boxed{\sum_{k=0}^\infty \left(\frac{1}{k!} \int_0^\infty x e^{-3x} \ln^k x dx \right) \varepsilon^k}$$

- (e) Use your results from parts (c) and (d) along with those from problem 1 to determine the value of the integral $\int_0^\infty e^{-3x} x \ln^2 x dx$. Your result should be written in terms of the irrational number γ , π , and $\ln 3$ only, along with rational numbers.

$$\int_0^\infty e^{-3x} x \ln^2 x dx = 2!c_2 = \boxed{\frac{1}{9} \left(\frac{\pi^2}{6} + (\gamma + \ln 3)^2 - 2\gamma - 2\ln 3 \right)}$$

3. Use the integral expression for the Beta function derived in class to determine the value of the following integrals. You may include π , $\sqrt{3}$, $\Gamma(\frac{1}{3})$, and $\Gamma(\frac{1}{4})$ in your expressions, but no other numbers (note that this means $\Gamma^2(\frac{1}{3})$ is allowable, but $\Gamma(\frac{2}{3})$ and γ are not).

(a)

$$\int_0^1 \sqrt{\frac{t}{1-t}} dt = \int_0^1 t^{\frac{1}{2}} (1-t)^{-\frac{1}{2}} \frac{\Gamma(\frac{3}{2})\Gamma(\frac{1}{2})}{\Gamma(2)} = \boxed{\frac{\pi}{2}}$$

(b)

$$\int_0^1 t^4 \sqrt{1-t^2} dt, u = t^2 \xrightarrow{u=t^2, t=\sqrt{u}} \int_0^1 u^{\frac{3}{2}} (1-u)^{\frac{1}{2}} du = \frac{\Gamma(\frac{5}{2})\Gamma(\frac{3}{2})}{2\Gamma(4)} = \boxed{\frac{\pi}{32}}$$

(c)

$$\int_0^1 \sqrt[3]{t^2(1-t)} dt = \int_0^1 t^{\frac{2}{3}} (1-t)^{\frac{1}{3}} dt = \frac{\Gamma(\frac{5}{3})\Gamma(\frac{4}{3})}{\Gamma(3)} = \frac{1}{9} \Gamma(\frac{1}{3}) \Gamma(\frac{2}{3}) = \boxed{\frac{2\pi\sqrt{3}}{27}}$$